Title

Ventura Marsh Milk-vetch (Astragalus pycnostachyus var. lanosissimus) 2024 Management and Monitoring Report

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Data Availability

The data associated with this publication are available upon request.

Ventura Marsh Milk-vetch

(Astragalus pycnostachyus var. *lanosissimus)*

2024 Management and Monitoring Report

UC Santa Barbara North Campus Open Space
December 2024

Cheadle Center for Biodiversity and Ecological Restoration
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Introduction

UC Santa Barbara's Cheadle Center for Biodiversity and Ecological Restoration (Cheadle Center) began working with the US Fish and Wildlife Service to plan for the introduction of the federally-endangered Ventura marsh milk-vetch (*Astragalus pycnostachyus var. lanosissimus*) at UCSB's North Campus Open Space (NCOS) in 2018. The first individuals were planted in a sandy site with high sub-surface soil moisture on the upper edge of the highwater mark of the newly restored upper arm of Devereux Slough in November 2019. These individuals were grown from seed collected from the original rediscovered population in Oxnard courtesy of Mary Carroll, the senior ecologist managing the site. In an effort to protect plants from inundation during times when the slough is fully ponded and to mimic conditions that may be related to success at planted locations in Ventura, five east-west trending berms approximately 1.5 meters wide, 0.5 meters tall, and 10 meters long were constructed with a small tractor. These were also designed to test the effect of subtle differences between north and south facing slopes as was observed in Ventura. The orientation of the berms later appeared to be unimportant at this site. At the same time, 6 milk-vetch were outplanted in the EEM Swale, a willow woodland with sandy soils near an ephemeral stream. In November 2019, 237 ten-month-old plants, grown at the Cheadle Center nursery, were planted on the site – 231 in the primary site "sandy zone" and six in the EEM Swale. Subsequently, more adults were planted on adjacent sandy mounds in the sandy zone in September 2020 (55) and October 2020 (46) (Figure 1). The outplanting effort is detailed in previous reports (Stratton et al., 2021).

Experimental seeding history

a. Seeding on NCOS

In December 2020, an estimated five cups of unprocessed seed were collected from the main population of previously planted individuals and dispersed experimentally across different zones of NCOS that were seen as potentially suitable habitat for the expansion of this initial population (Figure 1). The seeds were spread adjacent to the original planting in the "sandy zone" on a northeast-facing slope and sandy plain just southeast of the main planted population on similarly exposed fine sandy soils. The "Tule Seep" is a perennially wet seep surrounded by arroyo willow *(Salix lasiolepis)* and tule *(Shoenoplectus californicus)* and underlain by clay and clay loam soils. Whittier Pond is a small freshwater pond that holds stormwater year round. The "bioswale" site is a string of vernal pools descending in elevation northwest from the mesa into the saltmarsh and has dense clay soils. Finally, several square meter plots were seeded on South Parcel on sandy soils, in mixed riparian woodland and coastal sage habitats. Each of these five zones received one cup of collected seed and fruit material. In November of 2023, a sixth seeding zone, "vernal pool swale," was added to NCOS, extending south upstream from the Bioswale population and encompassing a string of six established vernal pools. The outer transitional zone of each vernal pool was seeded with about one quart of fruiting material, for a total of ~6 quarts, collected in summer 2023. At each seeding site, germinating seedlings were observed and monitored on a monthly basis during the growing season in the first 3 years, and four times over the course of 2024.

b. Seeding Off NCOS

In the winter of 2022-23, an additional nine experimental plots were seeded at the adjacent Coal Oil Point Reserve (COPR) and two at the Campus Lagoon. Several previous attempts to establish populations at COPR were ultimately unsuccessful. Despite initial vigor, the out-planted individuals were lost to various factors, including long periods of inundation, low precipitation in dry years, rodent herbivory, and the colonization of the open planting sites by willows, ruderal natives, and creeping wild rye grass *(Elymus triticoides)* (USFWS, 2020). The 2022-23 seeding effort was paired with four index wells to study the relationship between milk-vetch success and the height of the water table and groundwater salinity. Soil samples were taken from each seeding site and analyzed for soil salinity, percent moisture, pH, and nutrient content. Findings were consistent with previous conceptions of the milk-vetch's life history. The only successful seeding sites were those with a high water table throughout the year. Milk-vetch tended to prefer saltier groundwater, low soil SAR, and sandy, well-drained, nutrient-poor soils. The full results are detailed in the previous annual report (Chapman et al., 2024b).

Despite initial vigor in several of the seeding sites in 2023, there was only one plot, near the Dune Pond, where milk-vetch reached reproductive maturity. The Dune Pond is a groundwater-fed fresh to brackish coastal pond on COPR. The adjacent experimental plot often floods during the rainy season; it was inundated for around six weeks during February and March of 2024. As a result, no milk-vetch from the previous year survived the winter senescence. In 2024, new germinates at the COPR and the Lagoon seeding sites were monitored four times between spring and fall. Water level and quality data were collected from the index wells.

Several volunteer populations and individuals were discovered in 2023 and 2024 far from any of the purposefully seeded or planted sites. These include a group of 36 seedlings on the mesa grassland ("Grassland"), 8 near some public benches by the parking lot ("Roost"), and a single individual on the bank of the slough's eastern arm near Dilling's Link bridge ("Dilling's Link"). We theorize that these seeds may have been dispersed via animal or water, but the exact mechanism is unclear. Nevertheless, the milk-vetch's ability to disperse unaided by humans implies a certain level of resilience and sustainability in the NCOS metapopulation.

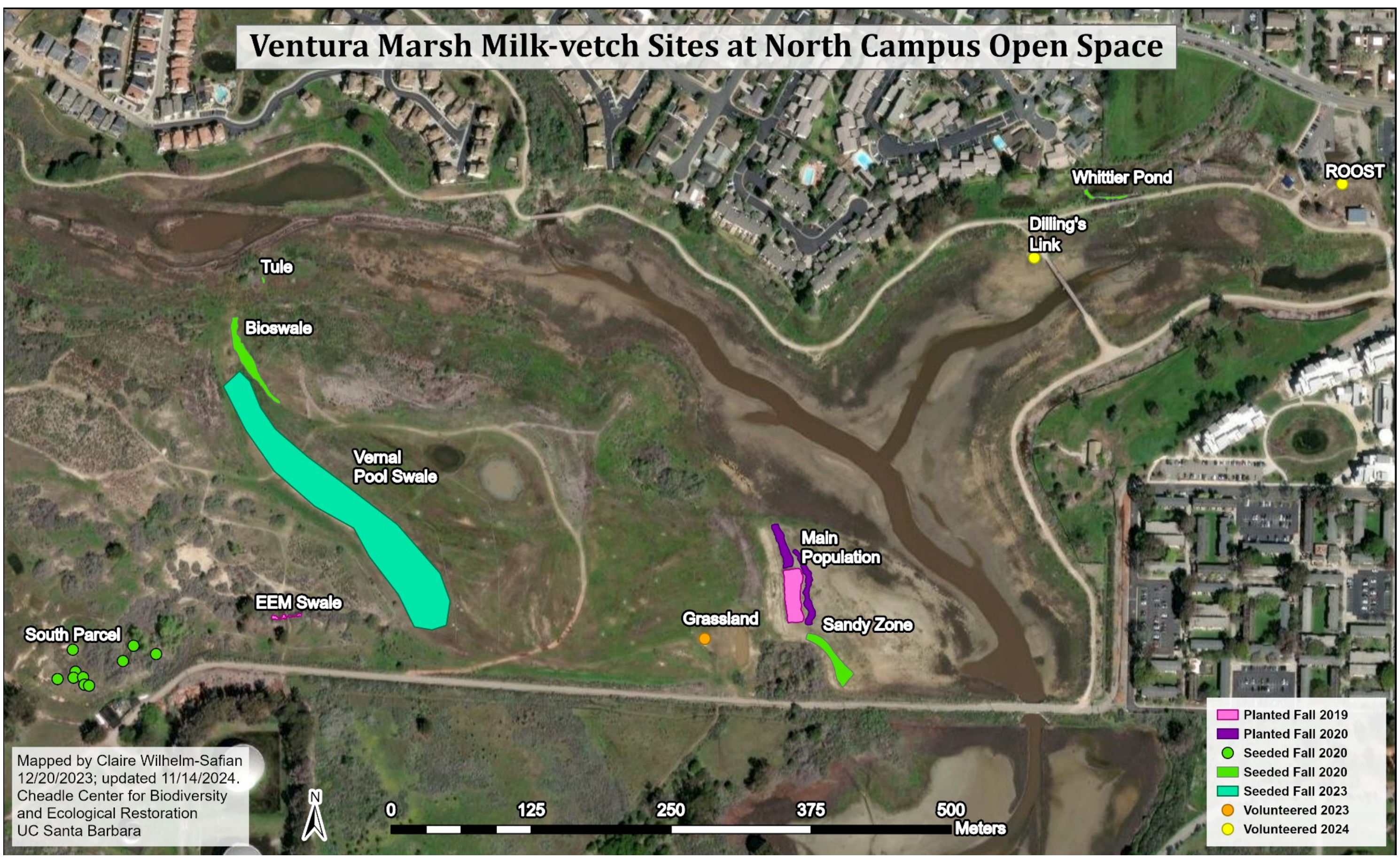

Figure 1: Map of milk-vetch plantings and seed dispersal sites at North Campus Open Space.

Main Population

The main population on NCOS was monitored in August 2024. Each adult was counted and its vigor was evaluated on a scale of 1 to 4 (1 = good, 2 = fair, 3 = poor, 4 = dead). The number of seedlings was estimated using a random sampling method stratified by population density. The entire seedling population was mapped and classified by seedling density class, defined as follows:

none = 0 seedlings/m^2
low density = 1-20 seedlings/m^2
medium density = 21-40 seedlings/m^2
high density = 41+ seedlings/m^2

In each class, three square meter quadrat samples were collected, for a total of 12 samples (Fig 2). The samples of each density class were averaged and multiplied by the percentage of total area assigned to each density class to estimate the total number of seedlings (Table 1, Fig 3). The total number of seedlings was estimated at 3,154; only 2,297 were alive at the time of surveying. The distribution of seedlings was patchier than in previous years, with a higher proportion of empty area (~83% in 2024, ~53% in 2023). Patches of densely clustered seedlings tended to be in poorer health (average vigor 3) compared to those with low or medium density (1.17 and 1.14). This may be due to intraspecies competition; in one high density sample, there were 40 seedlings, but only 6 were still alive.

Figure 2: Photo of a quadrat used to estimate the abundance of seedlings in the main population.

Density Class	Area (m2)	% of Total Area	Avg Seedlings/ m2	Average Vigor	Survivorship	Estimated Living Seedlings	Estimated Total
Low	106.63	12.16	7	1.17	95.2%	711	746
Medium	11.91	1.36	25.67	1.14	98.7%	302	306
High	26.73	3.05	78.67	3	61%	1,284	2,102
None	731.48	83.43	0			0	0
Total	876.75	100	27.83	1.77	72.77%	2,297	3,154

Table 1: Calculations for estimating seedlings in NCOS main population.

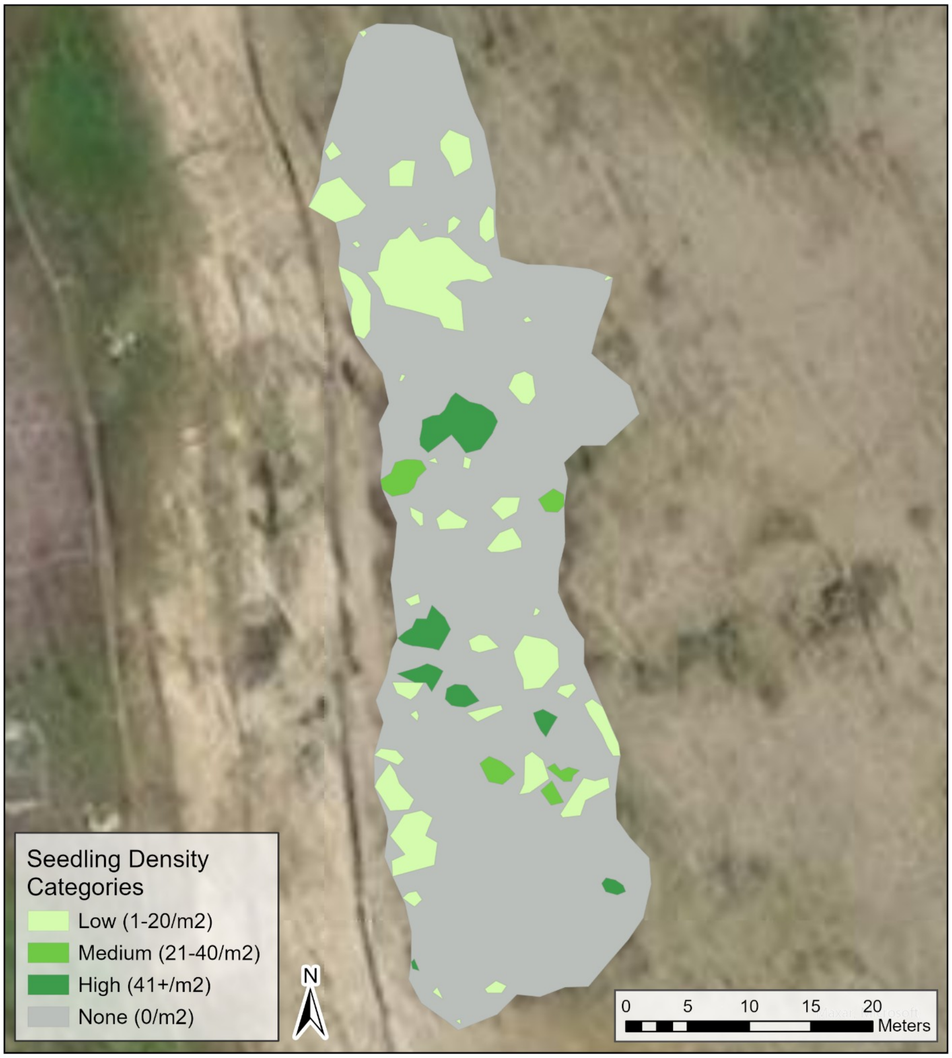

Figure 3: Map of seedling density classes used to estimate the size of the NCOS main population.

In previous years, each adult milk-vetch was tagged and tracked individually. However, due to the density and size of the population, it was logistically infeasible to maneuver within the area in 2023 without trampling and potentially killing plants. In that year, a sampling survey strategy was used to estimate the adult population (Chapman et al., 2024a). Although the population size and density declined in 2024, the continuity of that granular data had been disrupted, with many of the 2023 generation left untagged. In addition, due to erosion of the slope above the population, large amounts of sand have been deposited within its area, burying the ID tags from previous generations, or leaving it unclear which individual the tags referred to. Rather than tracking each adult individually, the main population was counted once in August, during the week of the seedling survey.

The total number of adults counted in the main population was 308. This is a significant decrease from the estimated adult population of 814 in the previous year. Only half (51.3%) were reproductive. In the previous year, the population was monitored two weeks earlier and still had a higher reproductive rate of 75.5%. Anecdotally, the bloom period appeared to be delayed this year, so the reproductive rate may have increased in September and October. Approximately one liter of seed and chaff material was collected in October from both the main population and the adjacent sandy zone for use in future conservation efforts.

Due to funding limitations, active management of invasive and native competitors within the main population was less frequent than in previous years. In addition, two back-to-back wet years spurred an increase in weedy non-native cover across the entirety of NCOS. Non-native species, largely dominated by *Polypogon monspeliensis* and *Melilotus indicus* (also including *Festuca myuros, F. perennis, Bromus hordeaceus, B. tectorum, Plantago coronopus,* and *Pseudognaphalium luteoalbum)* were removed by hand in spring. An increase in milk-vetch germination and growth was observed following removal. By summer, much of the empty sand cleared of weeds was replaced with native forbs such as *Heterotheca grandiflora, Erigeron canadensis, Epilobium ciliatum, Ambrosia psilostachya*. On the eastern edge, where there is a slightly higher water table, several patches of wetland plants such as *Juncus bufonius, Bolboschoenus maritimus, Distichlis spicata*, *Salix sp.,* and *Cyperus eragrostis* have expanded their footprints.

Inversely, the footprint of the milk-vetch population has shrunk from 1146 m^2 in 2023 to just 877 m^2 in 2024. Many of the individuals on the northern and southern edges, which were minimally managed in the previous year, died off in the winter of 2023-24. What was once open, unvegetated sand when the milk-vetch was first planted in fall of 2019 has gradually been colonized by other species of forbs and grasses. The milk-vetch is a nitrogen-fixing plant, so its presence on the landscape has made it possible for other species to gain a foothold in this sandy plain. Unfortunately, one of those species is *Melilotus indicus,* itself an N-fixing legume. Excessive nitrogen-fixation may represent a positive feedback loop that exacerbates invasion and competition. One should also take into account the milk-vetch's short life expectancy of 3-5 years (though according to managers at McGrath State Beach, a few individuals as old as 10 years have been recorded in Oxnard). Considering that the first generation of plants were outplanted in this zone in Fall of 2019, this population decline may represent the natural ecological succession of the plant community. Fewer active management work hours, increasing plant-available nitrogen, two successive wet rain years, and an increasing seed bank have all played a role in the increase of weedy species and the decline of Ventura marsh milk-vetch.

Other NCOS Site Monitoring

As with the main population, milk-vetch abundance dropped from the previous year's in every other seeding/planting zone on NCOS (Fig 4). Overall, recruitment was not high enough to match the mortality rate of older generations. Just as in previous years, no milk-vetch seedlings germinated at Whittier Pond. While a handful of seedlings germinated at South Parcel in earlier years, none were observed in 2024. These two sites were only monitored twice in March and in July. The edge of Whittier Pond was relatively open and sparsely vegetated in 2020, when the milk-vetch seed was sown, but has since filled in with thick tule *(S. californicus)*, California wild rose *(Rosa californica),* and arroyo willows *(S. lasiolepis).* Similarly, the South Parcel plots have been overgrown with Spanish clover *(Acmispon americanus)* and annual non-native grasses such as soft brome *(Bromus hordeaceus).*

Monitoring Year	Zone: Main Population	EEM Swale	Sandy Zone	Tule	Bioswale	South Parcel	Whittier Pond	Vernal Pool Swale	Grassland	ROOST	Dilling's Link
2020	197	5									
2021	524	6	59	66	40	3	0				
2022	18,881	22	164	32	36	0	0				
2023	15,078	30	18,066	16	28	1	0		36		
2024	3,462	0	960	15	9	0	0	4	21	8	1

Total Abundance
0 18,881

Figure 4: Table showing the yearly abundance of milk-vetch across all North Campus Open Space sites.

Monitoring Year	Main Population	EEM Swale	Sandy Zone	Tule	Bioswale	South Parcel	Whittier Pond	Vernal Pool Swale	Grassland	ROOST	Dilling's Link
	Zone										
2020	177	5									
2021	461	4	9	4	2	0	0				
2022	1,858	3	75	15	10	0	0				
2023	615	2	160	16	7	0	0		0		
2024	158	0	84	15	2	0	0	0	2	0	1

Reproductive Plants
0 1,858

Figure 5: Table showing the yearly number of flowering or fruiting plants at each North Campus Open Space site.

EEM Swale

This site was characterized by sandy to clay loam soils, an overstory of *S. lasiolepis*, and nearby well established *Eleocharis palustris* in the lower swale. *Salix* roots were heavily established in the planting sites, and it had more European garden snails than other sites due to the woody debris and cover. Originally planted with 6 adult nursery-grown milk-vetch in 2019, the EEM population saw some recruitment in the first few years, with a high of 30 individuals in 2023. However, the reproductive rate at this site was consistently low over the years; only between 2 and 5 individuals flowered each year (Fig 5). The population completely died off over the winter and no seedlings germinated in the spring to replace them. The swale stayed wet and muddy for much of the spring into summer. Rabbit's foot grass *(Polypogon monspeliensis)* heavily encroached the area, leaving little space for milk-vetch seedlings.

Sandy Zone

The sandy zone (referred to as "plover zone" in some previous reports) is a broad area originally consisting of mostly bare sand of fine grain similar to beach sand, with a stable level of sub-surface soil moisture. It was only monitored once in August, about a week after the main population survey. In 2023, a similar stratified random sampling method was used to estimate the number of seedlings, while each adult was counted. This year, it was apparent that germination had been much lower, so both adults and seedlings were counted. Indeed, milk-vetch abundance dropped from an estimated 18,066 in the previous year to 960 (Fig 6). At the time of the survey, 72 of these were dead seedlings. Of the 116 adults, only 84 were reproductive, a near 50% decrease from the previous year.

Monitoring Year	Seedlings	Adults	Flowering	Total Abundance
2021	50	9	9	59
2022	89	75	75	164
2023	17,885	181	160	18,066
2024	844	116	84	960

Figure 6: Table showing the population size of the sandy zone between 2021 and 2024.

Despite the drop in population size, the sandy zone was one of the healthiest populations this year, with an average vigor of 1.85 (Fig 7). Some of the older sandy zone adults near the main population died off, while younger, more vigorous adults with only minor health defects were clustered on the south-east slope. This geographic shift may indicate a change in soil or groundwater conditions on the site. See later discussion on the potential impacts of nitrogen-fixation.

Zone Id	
Vernal Pool Swale	4.00
ROOST	3.63
Grassland	3.62
Bioswale	3.22
Main Population	2.84
Sandy Zone	1.85
Tule	1.00

Average Vigor

1.00 4.00

Figure 7: Average vigor across all NCOS populations in August 2024, during peak bloom period.

Bioswale

The bioswale site is characterized by heavy clay to clay loam soils, some perennial native grasses, weedy annual grasses *(Festuca perennis)*, *Baccharis pilularis* and *Deinandra fasciculata*, and an eroding swale running through it. The population at the bioswale seeding site has dwindled from previous years. This area was largely overgrown with bur clover *(Medicago polymorpha)* and annual grasses due to the previous two wet winters. Seven adults from previous generations died by the end of the year, leaving only 2 milk-vetch alive (Fig 8). Both adults produced flowers and fruits, but with no recruitment of new seedlings this year, it remains to be seen whether this population will bounce back in 2025.

Vernal Pool Swale

The vernal pool swale seemed to be a good site candidate for seeding due to observations of the milk-vetch doing well in other locations around the site where it was seeded or recruited spontaneously, including upland locations in heavier soils that had no notable sub-surface soil moisture, such as the upper locations of the bioswale site and various volunteer locations. In fall of 2023, seed was spread along the vernal pools extending south of the bioswale population. Seeds were sown in the vernal pools themselves as well as the edges, due to the propensity of these pools to rapidly absorb water even on wet years. In 2024, only 4 seedlings were observed. They remained quite small, only a few inches tall, and none survived to the end of the year. The wet winter triggered a boom of bur clover in this area, which was treated multiple times with flame weeding, using high-intensity bursts of fire to wilt the germinating clover seedlings. Due to the thickness and abundance of the bur clover, it is possible that milk-vetch germinates were overlooked and burned in January or February, before they grew any true leaves. Outplanting a few nursery-grown milk-vetch may be a better method to determine whether the vernal pools are suitable habitat, and it may provide better habitat in drier years.

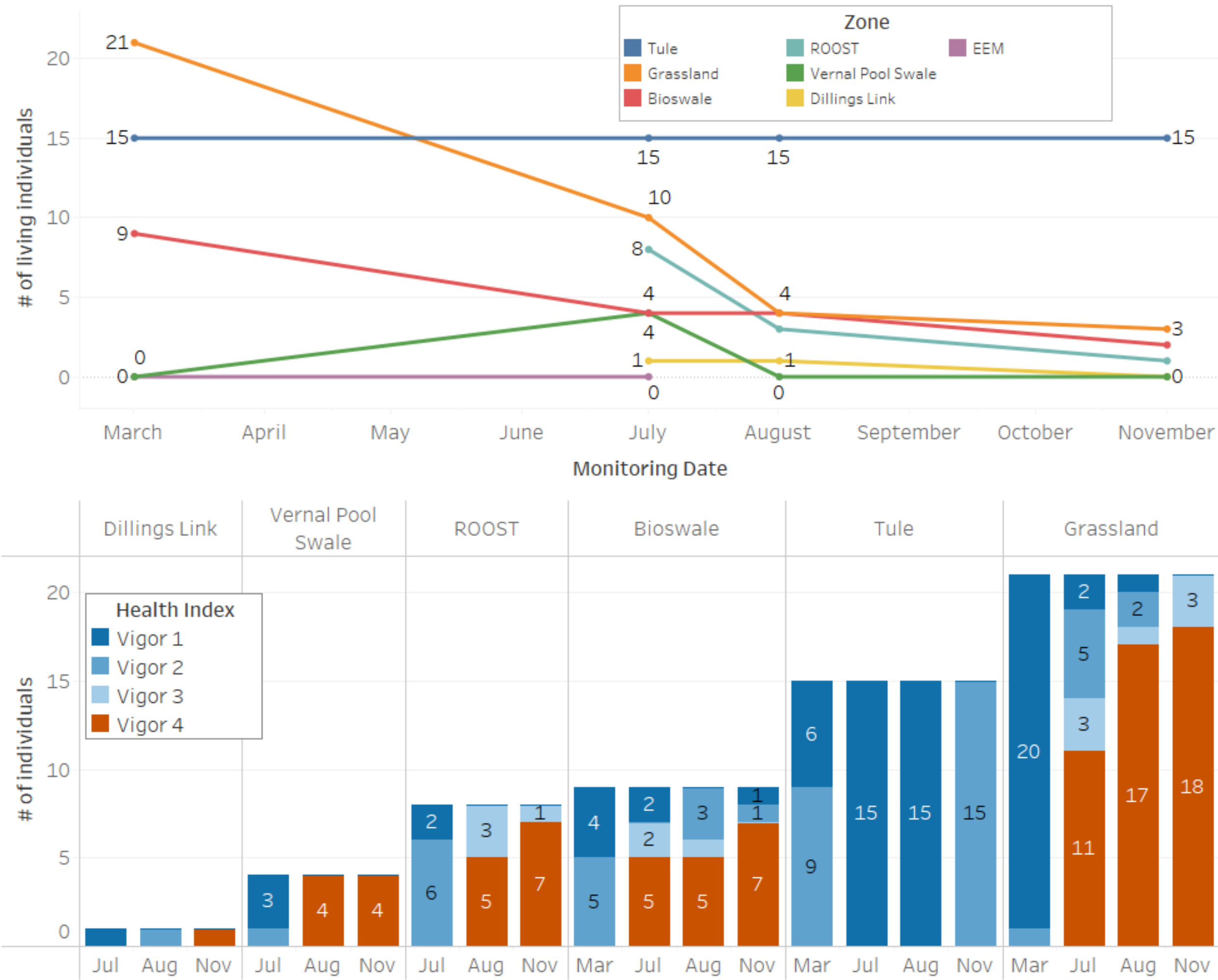

Figure 8: The population and health index of NCOS seeding populations over summer and fall of 2024.

Tule

The tule site is characterized by heavy sandy loam to clay loam soils, nearer the upper salt marsh edge, with some perennial sub-surface soil moisture that supports *Schoenoplectus*, *Salix*, and non native irises that remain from the golf course era. The population at the tule site has remained consistent since late 2022, with around 15 adults that flower and produce fruits each year. However, there has been no recruitment since the first year after seeding. This site experiences very little disturbance and is shaded by arroyo willows and tule. The milk-vetch here are larger and leggier than those in the main population and sandy zone, which receive full sun throughout the day. The site would probably benefit from some more aggressive weeding, disturbance, and the removal of the robust stand of irises.

Volunteer Populations

In the NCOS mesa grassland, a small group of milk-vetch seedlings were found in 2023, having germinated without any purposeful human dispersal. Though they remained small, 10 of these individuals survived the winter. There was no recruitment in 2024, though by the summer, two of them reached reproductive maturity and set seed. Two other individuals were observed closer to the top of the slope overlooking the main population.

A group of 8 seedlings were found in June of 2024 growing on a mound just outside the North Campus Open Space field lab, nicknamed the ROOST. This mound was created as part of the outdoor classroom and is largely vegetated with coastal sage scrub species such as *Artemisia californica* and *Encelia californica.* The soil is compacted clay and clay-loam. The seedlings remained small, with discolored gray leaves. Only 1 individual remained by the end of the year. The lack of groundwater most likely contributed to their demise. The site is half a kilometer from any other successful seeding site, and 70 meters from the slough itself. Animal or accidental human dispersal is the most likely mechanism.

Finally, a volunteer milk-vetch was found near Dillings' Link bridge, near the high water mark of the slough, most likely dispersed via water. This individual was originally discovered in July, and was already around 2 feet tall. While it did produce seeds in August, it displayed graying foliage throughout the year, declining in health and dying by November. The dead stems also appeared to have been chewed by a rodent or rabbit. This spot will be monitored in 2025 for any second generation seedlings.

Coal Oil Point Reserve/Campus Lagoon Monitoring

In an effort to determine suitable milk-vetch habitat at Coal Oil Point Reserve, 9 experimental plots were sown with milk-vetch seed in November 2022 and March 2023. In 2023, 623 seedlings germinated across seven of the plots (excluding plots 1 and 2 within the back dune habitat). By the end of the year, only two plots still had living milk-vetch: plots 6 West Mouth and 7 Dune Pond. Both of these plots were flooded after a rainstorm in mid-March, killing the existing vegetation and prompting a flush of germination from the milk-vetch seedbank. While the milk-vetch reacted to this stimulus at both sites, the population dwindled from 232 to just 9 individuals at plot 6, while 77 remained at the Dune Pond by the end of the year. There were 6 individuals that produced fruit at the Dune Pond, while none reached reproductive maturity at plot 6. Further details can be found in the 2023 experimental seeding report (Chapman et al., 2024b).

Germination was comparatively low at most sites in 2024, though there was a significant increase at plot 4 West Main Channel and a small increase at plot 5 West Main Channel (Fig 9). As previously mentioned, the only site with milk-vetch that reached reproductive maturity was 7 Dune Pond, so the vast majority of germination can be attributed to the seeds originally sown in the winter of 2022-23. The highest germination was observed at plot 8 East Main Channel, with a high of 79 individuals in March (Fig 10). However, this site was densely vegetated with grasses and forbs *(Bromus diandrus, Phacelia ramosissima, Camissoniopsis cheiranthifolia)*, and the small germinates struggled to compete. None survived to summer (Fig 11). There were no seedlings at plot 6 West Mouth when it was visited in March. The site had recently flooded; the sand was saturated and all of the vegetation was dead. By July, 39 milk-vetch seedlings had germinated in the cleared space. The population decreased to 11 by the end of the year. None of them flowered. Another 34 seedlings were observed at plot 4 West Main Channel. The vegetation is denser at this plot compared to most of the others, with competition from rabbit's foot grass *(P. monspeliensis),* alkali bulrush *(Bolboschoenus maritimus),* sowthistle *(Sonchus sp.),* poison oak *(Toxicodendron diversilobum),* and horseweed *(Erigeron canadensis)* (Fig 12). By the end of the year, 14 seedlings remained and none of them flowered. Twenty-nine seedlings were found at 5 West Main Channel in the spring, but the population dropped to 3 in July and they all died by mid-summer. Finally, the most successful plot of the previous year, 7 Dune Pond, only had 12 individuals in the summer, and dropped to 2 by November. None of the milk-vetch produced flowers or fruits, and the stems had been browsed by rodents or rabbits. This plot was overgrown with rabbit's foot grass *(P. monspeliensis)*, curly dock *(Rumex crispus),* and tall flatsedge *(Cyperus eragrostis)* (Fig 13). It is possible that while the flooding in March 2023 had cleared space for the milk-vetch, the area was subsequently colonized by these competitive weeds, additionally providing habitat for herbivorous rodents and rabbits.

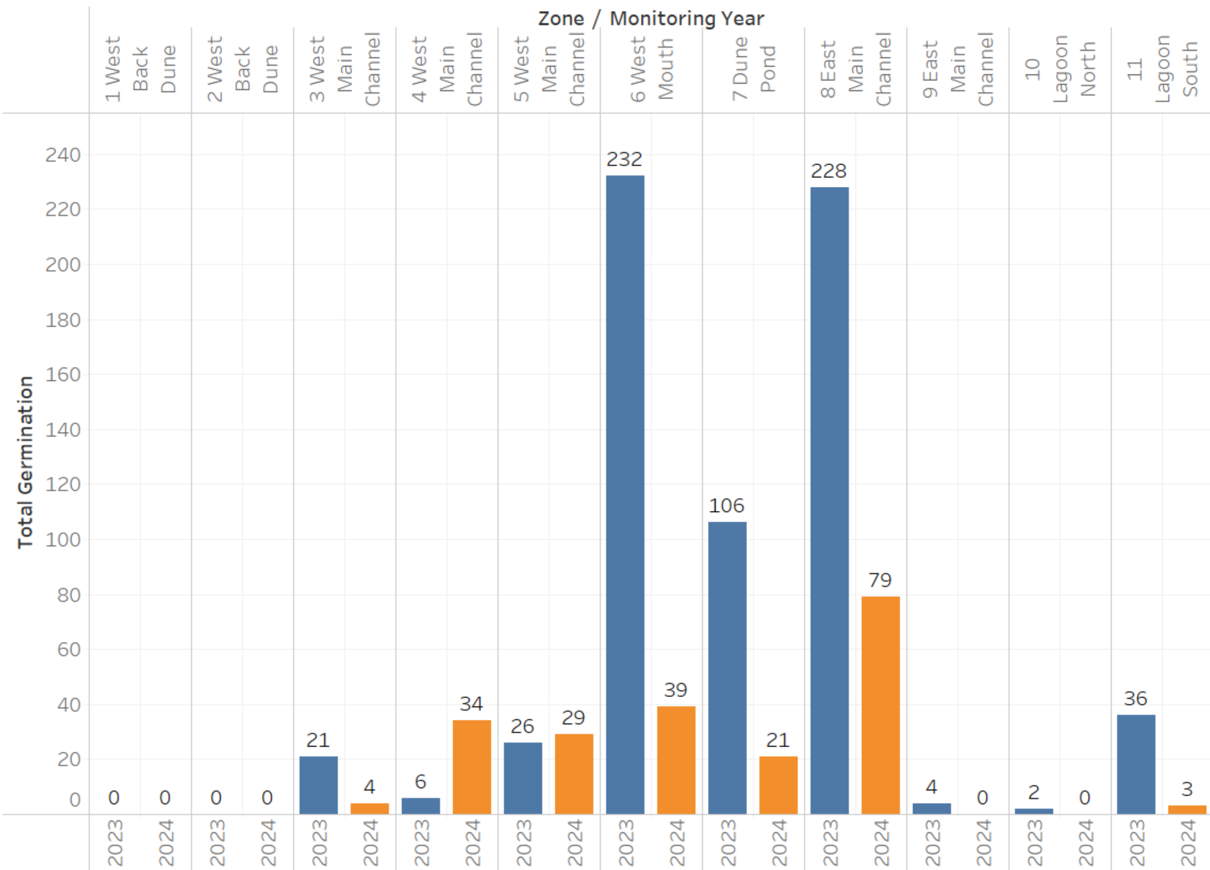

Figure 9: Bar graph comparing the yearly total germination across all COPR and Lagoon sites.

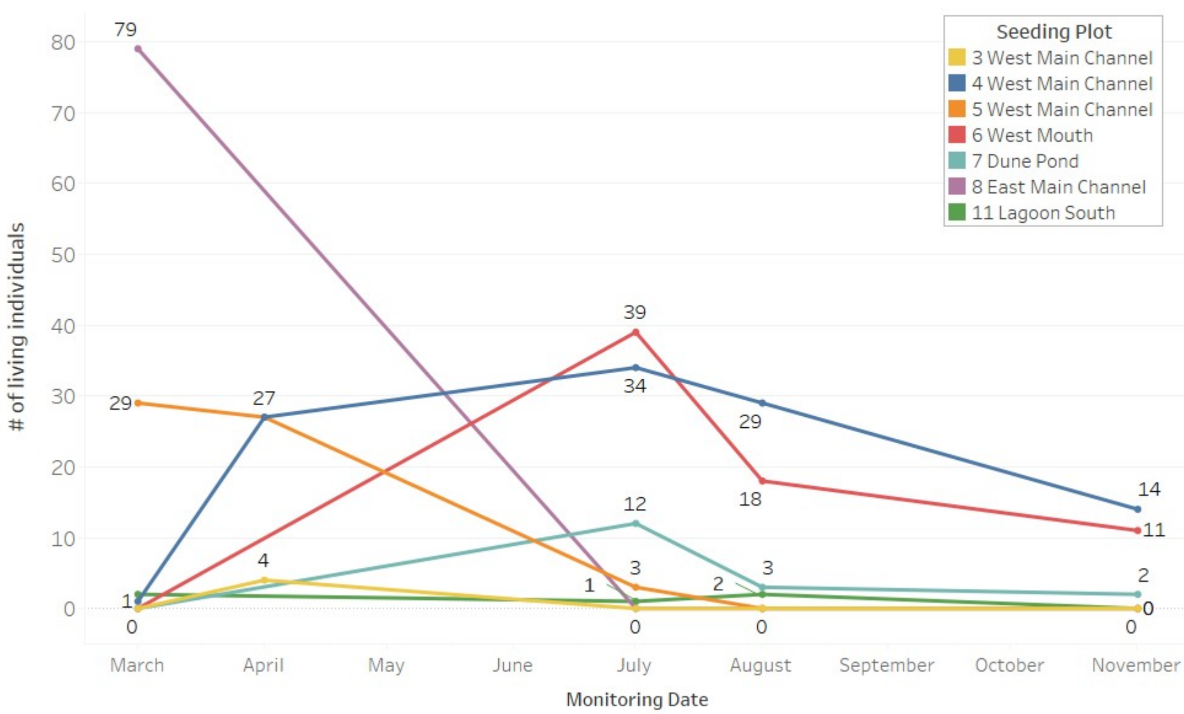

Figure 10: The abundance of living milk-vetch at COPR and Campus Lagoon seeding plots over time in 2024.

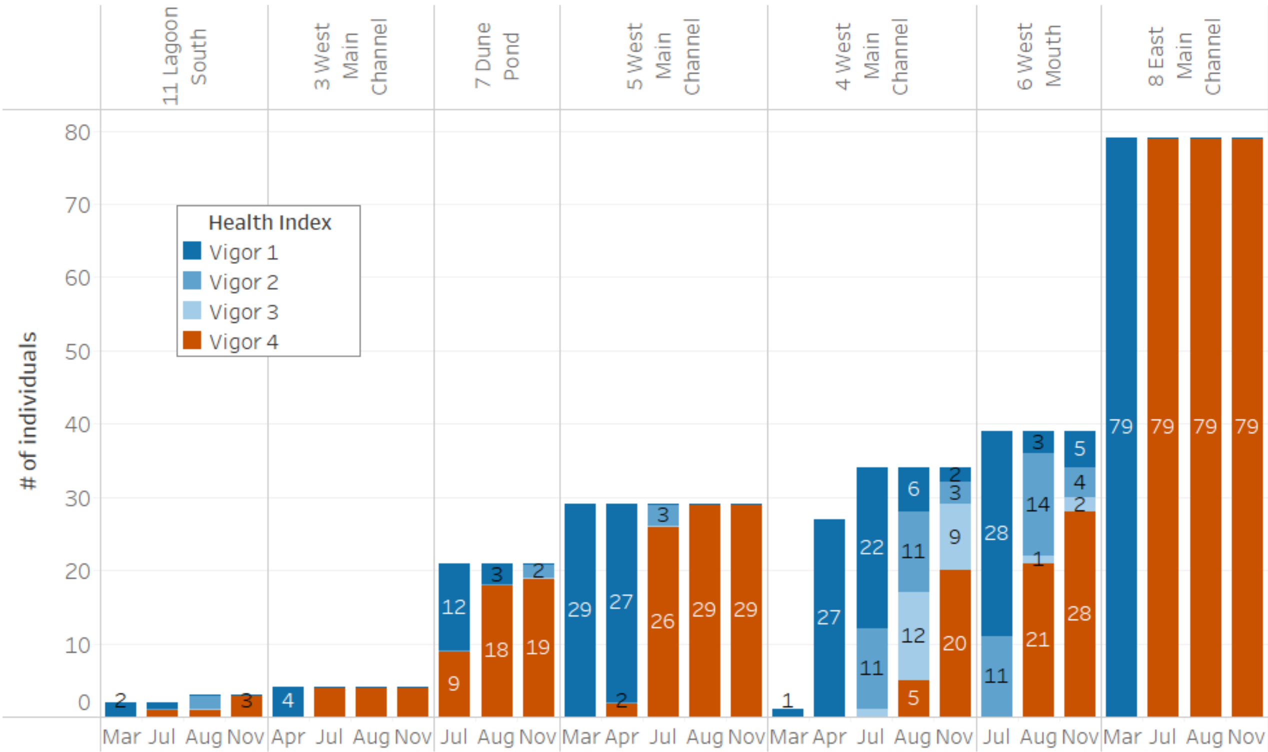

Figure 11: The vigor of milk-vetch at COPR and Campus Lagoon seeding plots over time in 2024.

Figure 12: A photo of plot 4 West Main Channel showing the dense competing vegetation in July 2024.

Figure 13: A photo of plot 7 Dune Pond showing the dense competing vegetation in July 2024.

In addition to those at Coal Oil Point, two plots (10 Lagoon North and 11 Lagoon South) were sown with milk-vetch seed in 2023. In 2024, no milk-vetch germinated at plot 10, and only 2 new seedlings germinated at plot 11. One individual from the previous year grew several feet tall, and flowered and fruited in the summer. This adult died back by November, though it may have been going into winter senescence.

Ventura marsh milk-vetch appears to do best with a high water table throughout the year, within 5 ft of the soil surface, although in heavier soils it seems able to persist without this to varying extents. Four 5-ft long index wells were installed at plots 5 West Main Channel, 7 Dune Pond, and 9 East Main Channel at Coal Oil Point Reserve and plot 11 Lagoon South at the Campus Lagoon. In 2024, the index wells at COPR were monitored with each visit. Each contained a pressure transducer (Solinst Levelogger® 5) hanging at the bottom of the well. The data was downloaded and compensated with data from a barometric pressure transducer to track the groundwater level over time. Upon each visit, the groundwater salinity (ppt) was measured using a YSI Pro2030 Meter. The well at 11 Lagoon South was regularly dry throughout 2023. During well installation,a layer of cobble was found around 4 feet below the soil surface, which may be increasing the drainage at this location, making it unsuitable for milk-vetch. The pressure transducer was removed from this well in early 2024 for use in other projects and the depth was instead measured by hand on each visit. The well continued to run dry throughout 2024.

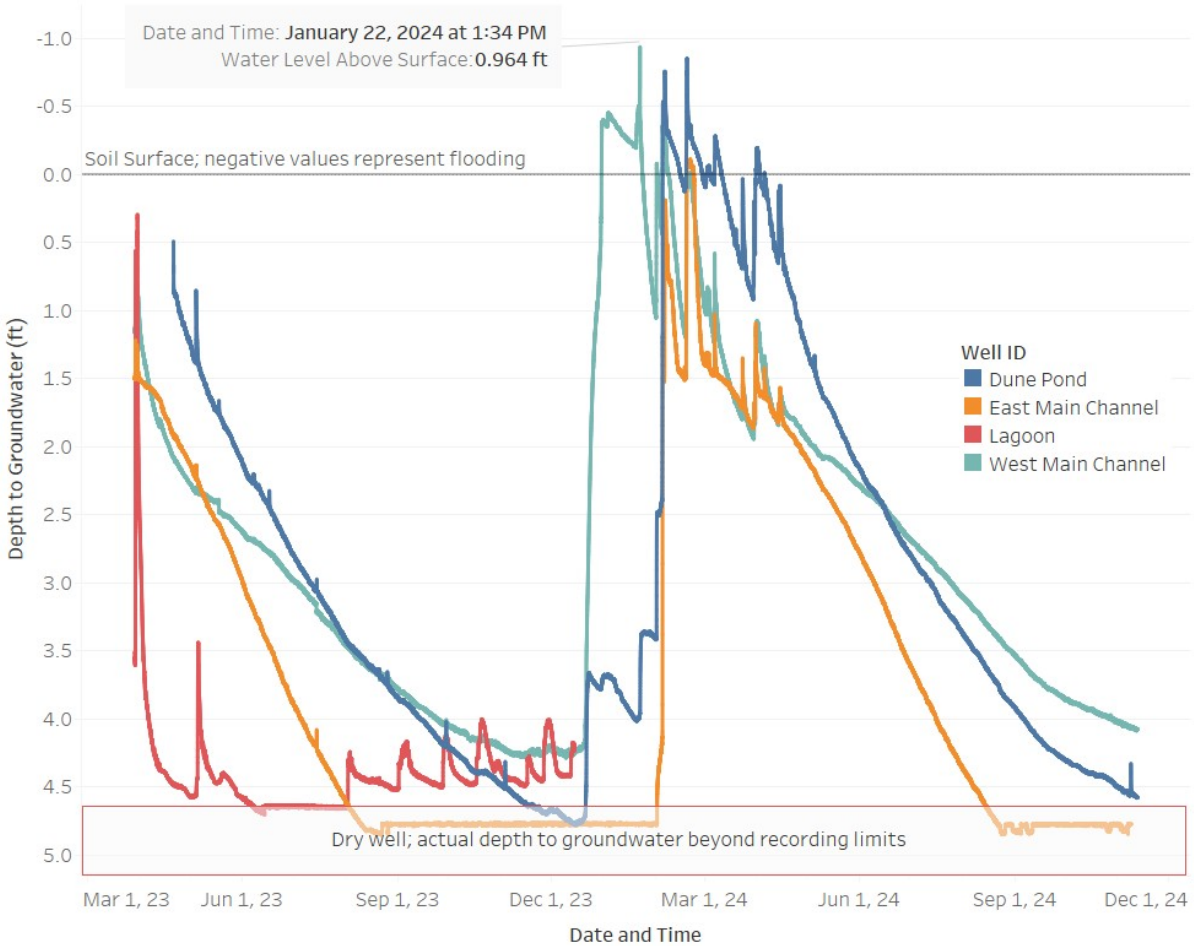

Figure 14: Line graph showing the groundwater level at each index well between March 2023 and November 2024, using data downloaded from the pressure transducers.

The wet winter appears to have recharged the groundwater at COPR. The water table was higher at the COPR wells in 2024 compared to the previous year (Fig 14). Each well site was flooded, though this was limited to a few days in February at East Main Channel. West Main Channel was ponded throughout January, before any milk-vetch would have likely germinated. However, the Dune Pond site was flooded between February 4 and March 13, with another shorter period of inundation during the first week of April. Cumulatively, these six weeks of flooding killed the previous generation of milk-vetch. The largest of these skeletons were observed once the water receded. While a second generation of seedlings did germinate, they struggled to compete with rabbit's foot grass, ribwort plantain, and curly dock, as previously discussed. While the high water table is ideal for the milk-vetch, if flooding continues to be a yearly occurrence, it is unlikely that this population will be sustainable in the long term. We recommend using satellite imagery to map flooding extent in previous years and identify an ideal outplanting or seeding site on the edge of this ephemeral pond.

A linear regression model indicates that depth to groundwater and salinity have a significant impact on the overall health of milk-vetch, though not necessarily its abundance (Fig 15). The population size increased with groundwater salinity and depth to groundwater (Fig 16). The vigor score ranges from 1 (healthy) to 4 (dead); thus, a low average score indicates good overall health, and a high average score indicates poor health. Depth to groundwater was strongly correlated with average vigor ($r^2 = 0.2765$, $p < 0.0119$), suggesting that as the water table lowered and the milk-vetch had less access to water, the population had more health defects and started dying off. Salinity was negatively correlated with average vigor ($r^2 = 0.3065$, $p < 0.0075$), indicating that the milk-vetch was healthier at sites with slightly saltier groundwater, in the range of 0.6-1.3 ppt.

Row	Column	DF	Term	Value	StdErr	t-value	p-value	r-squared
Living Population	Salinity	20	Salinity	26.588	13.95	1.9062	0.0711	0.1537
			intercept	3.185	10.73	0.2969	0.7696	
Living Population	Depth to Groundwater	20	Depth to Groundwater	5.1005	6.691	0.7623	0.4548	0.0282
			intercept	5.0356	20.2	0.2493	0.8057	
Average Vigor	Salinity (ppt)	20	Salinity (ppt)	-1.575	0.53	-2.973	0.0075	0.3065
			intercept	3.4554	0.407	8.4849	< 0.0001	
Average Vigor	Depth to Groundwater	20	Depth to Groundwater	0.6695	0.242	2.765	0.0119	0.2765
			intercept	0.5905	0.731	0.8077	0.4287	

Figure 15: A table of statistical values of the linear regression models of various groundwater quality measurements with average vigor and population size.

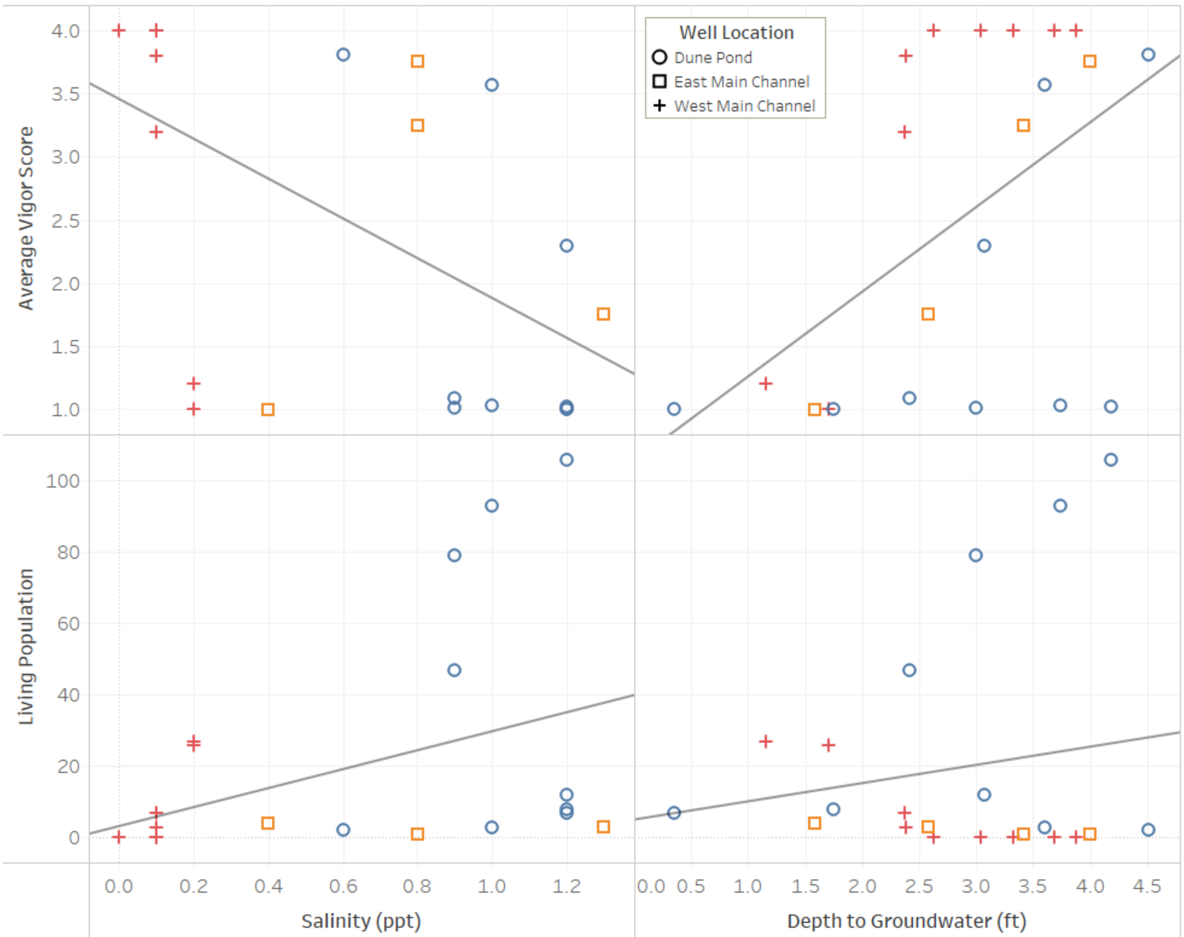

Figure 16: A grid of scatterplots showing the relationships between groundwater salinity and depth, versus milk-vetch population size and average vigor.

The Ventura marsh milkvetch appears to do best in stable sites with more regular sub-surface soil moisture, as it is vulnerable to both flooding and drought, while stochastic floods likely played a role in its dispersal and population dynamics in the past. COPR seems to be more susceptible to the stochasticity of weather patterns, drying down in dry years due to its sandier soils, and both flooding and supporting expansive growth of other plants during wetter years, such as *P. monspeliensis,* all factors that complicate the establishment and persistence of the Ventura marsh milkvetch.

Photo Monitoring

Standard photo monitoring points were established around the main population in October 2021 in order to capture visual changes of the population over time. Photos are usually taken once a year during the peak bloom. In 2024, photos were taken in late May. A selection of these photos are included in this report. The full photo monitoring dataset is available upon request.

Figure 17: Photo monitoring point 1A capturing the main population from SW corner (August 6, 2021).

Figure 18: Photo monitoring point 1A capturing the main population from SW corner (August 3, 2023).

Figure 19: Photo monitoring point 1A capturing the main population from SW corner (May 21, 2024).

Threats and Future Management Plans

The expansion of invasive non-native annual weeds including both rabbit's-foot grass (*Polypogon monspeliensis*), Italian ryegrass *(Festuca perennis),* and sweet clover (*Melilotus indicus*) has been a significant threat to the vetch over all sites. Both *M. indicus* and the milk-vetch are N-fixing plants, allowing *P. monspeliensis* to take advantage of the high N concentration in soil and take over the site. While extremely competitive, preliminary research indicates that *P. monspeliensis* is also allelopathic, and although the exact extent of this and the mechanisms are not well understood it may inhibit the germination, growth and survival of *A. pycnostachyus* ssp. *lanosissimus.* Further observations indicate that the milk-vetch may not prefer or survive well in N-rich soils, which is especially problematic given the limited footprints of ideal soils and soil moisture on the site, and its propensity to fix nitrogen. One possible factor here appears to be increasing leaf browning, seedling mortality, stunting of growth, reduced vigor, and reduced flowering or fruiting in soils that were previously supporting large, robust plants. A possible explanation for this are spider mites, which have been observed on the plants in the past, and may be worsening through time. Spider mites are known to be especially problematic on plants in soils with high levels of nitrogen. It is possible that the N-fixing abilities of the vetch and its ability to thrive in soils deficient in this important nutrient indicate a strong preference for soils low in nitrogen.

While the entire sandy zone was mostly devoid of vegetation at the site's inception, it has slowly accrued more and more vegetation, both native and non-native. Both the native Ventura marsh milk-vetch and non native legumes have been steadily adding plant-available nitrogen to the soil, and a subsequent increase in sickle grass *(Parapholis incurva),* rabbit's foot grass *(P. monspeliensis),* and salt grass *(D. spicata),* in addition to many other broadleaf taxa have been establishing. One notable observation was the increased height, size, and fecundity of both *Polypogon* and cut-leaf plantain *(P. coronopus)* around the bases of individual *Astragalus* across the site where milk-vetch grew for one or more years, and were killed by flooding (Fig 20). While surrounding areas maintain both of the non-native species at around 2-4" in height quite uniformly, the immediate area around the bases of former milk-vetches supported heights of both weeds exceeding 24". In some cases this appears to be clearly associated with *Astragalus,* while in others it may be associated with piles of *Melilotus* that were hand-pulled and left in place the previous year. Regardless, N-fixation and the associated growth, mostly of weeds, continues to be a serious and worsening problem.

In an effort to better understand this mechanism and the volume of N in the soil, a nitrogen study was conducted. Samples of *P. monspeliensis* were taken from three different substrate type zones: *Melilotus indicus* piles, milk-vetch, and other vegetation. From each zone, a sample of *P. monspeliensis* was collected and dried in the drying oven. The plant samples were separated by stems, leaves, and heads to gather proper information on the CHN content of each independent plant part. Once dried and separated, *P. monspeliensis* samples were finely ground into a powder and sent to the MSI analytical laboratory for CHN elemental analysis as well as delta N analysis. This study is still in progress, and while CHN data has been determined, isotope data has not yet been returned from the lab. To determine the biomass of both the milk-vetch, and other vegetation zones, samples of equal area were collected and dried in the drying oven before being massed. Five samples were taken from each zone and averaged to determine the mean biomass of each substrate type zone.

Figure 20: Two photos showing increased biomass of weeds growing around the bases of dead milk-vetch or spots where sweet clover had been piled the previous year, emphasizing the effects of nitrogen fixation and the associated problematic habitat conversion it creates.

Based on the nitrogen study data collected so far, the total nitrogen per unit area within *Polypogon* with nitrogen additions from the milk-vetch appears to be higher than the total nitrogen per unit area without (Fig 21). This indicates that although the *P. monspeliensis* is limited in the amount of nitrogen it is able to retain per unit mass, added nitrogen in soil allows *P. monspeliensis* to grow much larger, ultimately competing with the milk-vetch by shading it out, competing for root space, possible allelopathic effects, and generally overtaking its habitat. Additionally, the trend of stems holding the least amount of nitrogen and heads holding the greatest amount of nitrogen only in the *Polypogon* samples collected from the VMMV piles indicates that different sectors of the plant may hold significantly different amounts of nitrogen in VMMV zones. The idea that different amounts of nitrogen are held in each part of the plant is further supported by the relative ratios of nitrogen percentage in leaves to stems and heads to stems across substrate type zones in which the ratio is much closer to one in low nitrogen vegetations (Fig 22). This implies that *Polypogon* may be leafier and have more flowers in the milk-vetch substrate zone than in other areas of vegetation. However, the biomass data collected was measured from the plant as a whole, and not divided by heads, leaves, and stems, which leaves space for more investigation into whether or not this is actually the case.

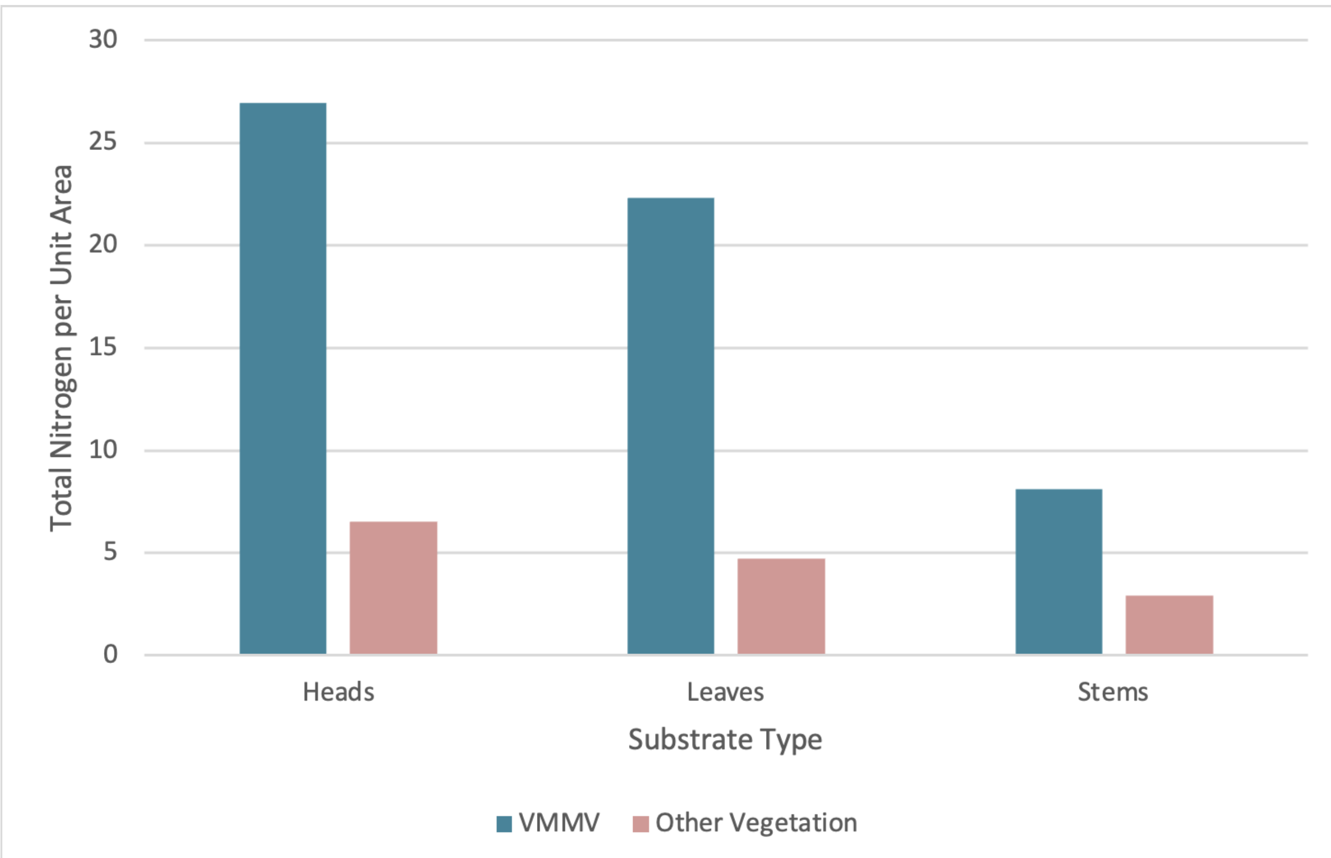

Figure 21: Total nitrogen per unit area with and without nitrogen additions from milk-vetch for heads, leaves, and stems.

Treatment	Leaf % N : Stem % N	Head % N : Stem % N
VMMV Zone	2.75	3.31
Other Vegetation	1.61	2.23

Figure 22: A table indicating relative ratios of percent nitrogen within leaves and heads to stems in both VMMV and other vegetation zones.

In order to manage the higher nitrogen content, a tractor was used to move soil from the main population and replace it with comparatively less N-rich sand. The northern end of the main population was chosen to test this management strategy. In 2023, this section was minimally managed and used as a control area to test treatments such as removing dead milk-vetch branches and applying graminicide on the rest of the main population. As a result, the area was comparatively overgrown with *P. monspeliensis, F. perennis, E. canadensis, A. psilostachya,* and *P. coronopus.* Although it had supported milk-vetch in previous years, there were no individuals growing there in 2024. In late November and early December of 2024, the top 0.5-1 foot of soil was removed from the northern end of the main population using a tractor (Fig 22). A roughly equal amount of soil was deposited in its place, taken from a section of mostly open sand with scattered saltgrass adjacent to the population. The aim is to remove some of the weedy seedbank and provide fresh, relatively nitrogen-poor soil for the milk-vetch. Monitoring will continue in spring of 2025 to track the efficacy of this strategy.

Figures 23-25: Photos of the soil removal from the northern section of the main population, recreation of lower N berms, and further creation in the middle of the population. Work is ongoing.

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